GNSS-Aided Semantic Landmark Particle Filtering for Row-Level Localisation in Vineyards

Anonymous Authors

***Abstract*— Global robot localisation in orchards and vineyards is hindered by row-level perceptual aliasing: parallel crop rows produce geometrically identical LiDAR and visual observations, causing SLAM and Monte Carlo localisation to settle in false adjacent corridors. In addition, foliage canopies induce severe GNSS multipath, drift, and signal outages. We present a Semantic Landmark Particle Filter (SLPF) that fuses 2D LiDAR scans, YOLOv9 semantic detections of stable trunks and support poles, and a noisy GNSS prior. Rather than treating landmarks as disconnected points, our method groups them into surveyed "semantic walls" formed by finite row-aligned segments, supplying local corridor constraints within the particle score. We formulate a Huber penalty for GNSS innovations and a penalty cap for semantic observations to bound outlier effects. Evaluated on two complete traversals of one vineyard over three random seeds $(11, 22, 33)$ against seven baseline configurations, SLPF achieves aligned APE below $1\,\mathrm{m}$ ($0.89 \pm 0.04\,\mathrm{m}$ and $0.91 \pm 0.03\,\mathrm{m}$) and raw map-frame APE below $1\,\mathrm{m}$ ($0.94 \pm 0.06\,\mathrm{m}$ and $0.98 \pm 0.08\,\mathrm{m}$), with in-row correct fractions of 0.71 ± 0.02 and 0.64 ± 0.05. These $n = 3$ summaries are descriptive; the canonical baseline matrix contains both dedicated and historical reference runs. We also report controlled GNSS-outage, landmark-removal, and execution-profile experiments.**

I. Introduction

Autonomous ground vehicles in viticulture enable precision spraying, mechanical canopy pruning, yield estimation, and harvesting [1], [2], [3]. However, long-term field operation requires continuous global pose estimation under severe perceptual ambiguity.

Vineyards exhibit pronounced *perceptual aliasing*: rows feature parallel, uniformly spaced trellises, identical post spacings, and repetitive vine foliage [4], [5], [6]. Standard 2D and 3D LiDARs observe nearly identical cross-sectional point profiles in every corridor. Consequently, geometry-only Monte Carlo localisation methods like AMCL [7] suffer from multi-modal dispersion and can collapse onto incorrect adjacent rows. Visual SLAM frameworks [8], [9] similarly produce false loop closures between distinct parallel rows or lose tracking during lighting shifts and direct solar glare.

Although Global Navigation Satellite Systems (GNSS) provide absolute world-frame positioning, high-precision Real-Time Kinematic (RTK) signals are frequently degraded by canopy multipath, satellite occlusion along valley slopes, and foliage attenuation [10], [11], [12]. Standard GNSS receivers exhibit position errors exceeding $3.0\,\mathrm{m}$, larger than typical vineyard row widths (2.0–$2.8\,\mathrm{m}$), making raw GNSS alone incapable of resolving row identity (Fig. 1). This limitation is acute at headland turning zones, where the robot exits crop rows, executes rapid yaw transitions, and must enter the correct target corridor.

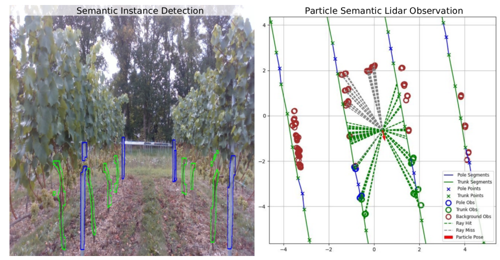

Fig. 1: Projection of detected semantic landmarks (vine trunks and support poles) onto the surveyed vineyard map. The proposed SLPF groups adjacent landmarks into finite row-aligned semantic-wall segments to address corridor aliasing.

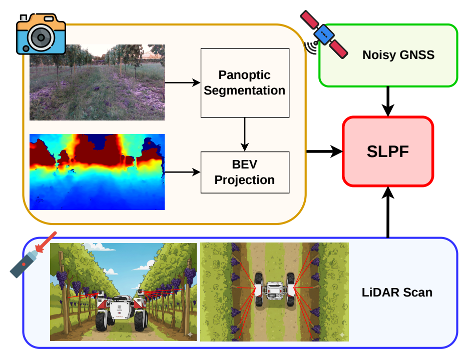

Fig. 2: Overview of the Semantic Landmark Particle Filter (SLPF) architecture. Wheel odometry drives particle propagation, while 2D LiDAR scans, YOLOv9 semantic detections, and Huber-gated GNSS measurements update particle weights. The figure's "Panoptic Segmentation" block denotes YOLOv9 instance-mask detection, and "BEV Projection" denotes depth back-projection followed by LiDAR association.

We address these challenges with the Semantic Landmark Particle Filter (SLPF). While seasonal foliage and weather alter visual appearance, structural woody vine trunks and rigid support poles remain stationary across seasons. We extract these vertical elements using an object detector and project them into the robot frame. Rather than treating detections as unordered points, we construct a structured *semantic-wall* representation of finite, class-labelled segments and the orientation of each surveyed row.

Within the particle filter, we combine odometry propagation, LiDAR-associated semantic-wall scores, a row-segment corridor score, and a Huber-attenuated GNSS score. The particle set represents pose alternatives during ambiguous transitions; row identity is evaluated using a separate nearest-row rule restricted to in-row frames.

We also resolve a common evaluation pitfall in agricultural robotics: conflating global map-frame error, rigid trajectory alignment, and operational row identity. Aligned Absolute Pose Error (APE) applies a global rigid transform that can mask genuine map-frame drift. Similarly, evaluating row correctness globally across all frames skews metrics at headlands where nearest-row assignments are ill-defined. We decouple map-frame raw APE, aligned APE, in-row corridor identity, and headland turning recovery.

The primary contributions are:

1) Semantic-wall formulation: A finite-segment measurement score mapping visual trunk and pole detections into row-aligned structural constraints, combined with a Huber GNSS model and a corridor prior.
2) Disentangled evaluation protocol: A metric protocol evaluating raw map-frame error alongside aligned APE, restricting row-identity metrics to in-row frames, and quantifying headland transitions via cross-track departure and recovery distance.
3) Protocol-bounded baseline benchmarking: Evaluation over multiple seeds and traversals against seven baselines (AMCL, NoisyGNSS, RTAB-Map RGB/RGB-D), eight controlled GNSS degradation profiles, and landmark decimation tests.
4) Component ablation: An eight-variant ablation describing measured changes in the registered configurations and releasing the execution protocol used to generate the results.

II. Related Work

A. Agricultural Robot Localisation and Particle Filtering

Particle filtering (Monte Carlo Localisation) is widely adopted in field robotics because it represents arbitrary non-Gaussian distributions arising from wheel slip, uneven terrain, and environmental symmetry [1], [3], [7]. In row crops, prior work has fused wheel odometry, IMU, and 2D/3D LiDAR scan matching [13], [14]. AMCL [7] uses a likelihood field or beam model over 2D occupancy grids. However, because neighbouring vine rows possess nearly identical geometric profiles, geometric particle filters in vineyards frequently suffer particle depletion or converge to parallel incorrect rows.

To mitigate drift, GNSS is often integrated into Kalman filters or factor graphs [12], [15], [16]. Yet multipath reflections from wet foliage and trellis wires introduce correlated non-Gaussian biases [10]. While M-estimators and Huber penalties have been applied in urban driving, combining them with row-structured semantic priors under agricultural GNSS degradation remains under-explored.

B. Visual and LiDAR SLAM in Repetitive Crops

Simultaneous Localisation and Mapping (SLAM) has been investigated extensively for orchards and vineyards [4], [6], [8]. Feature-based visual SLAM methods such as RTAB-Map [8] extract keypoints from canopy textures. However, repetitive foliage causes perceptual aliasing during loop closure, matching images across distinct rows that appear identical [4], [17]. Visual-inertial and visual-depth SLAM

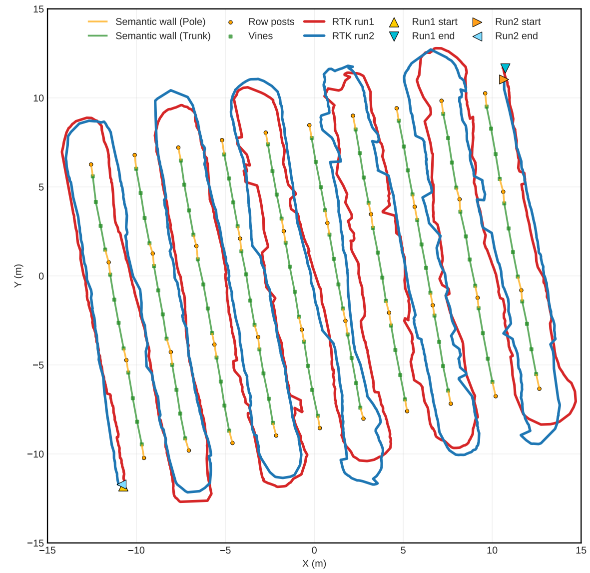

Fig. 3: Vineyard structural map with RTK ground truth for `rh_run1` (red) and `rh_run2` (blue). Circles and squares denote support posts and vine trunks; lines denote semantic walls. Triangles indicate start/end points.

maintain local smoothness but accumulate drift over long rows lacking distinctive landmarks.

Recent work investigated semantic keypoints and stable 3D points to enhance visual feature matching [5], [18]. Similarly, Aguiar et al. [1] extracted semi-planes and pole features from 3D LiDAR. While these approaches improve front-end tracking, they do not incorporate prior digital vineyard layouts to guarantee row-level consistency and require 3D point cloud pipelines that challenge constrained field platforms.

C. Semantic Landmark and Prior-Map Navigation

Semantic landmarks such as tree trunks, trellises, and support posts provide stable structural references for robot navigation [19], [20], [21], [22]. In orchards, Costley and Christensen [21] demonstrated landmark-aided navigation using tree trunks extracted from LiDAR ranges. Prior topological and landmark maps can also be acquired from low-altitude aerial surveys prior to ground rover deployment [23]. In urban driving, pole-like structures provide viewpoint-invariant features for particle-filter localization [19], [20].

Our proposed SLPF builds upon these foundations: (i) we group adjacent trunk and post detections within each surveyed row into finite class-labelled semantic-wall segments rather than independent point constraints; (ii) we incorporate a Huber GNSS score that reduces the influence of large innovations; and (iii) we establish a protocol separating in-row and headland metrics.

III. Semantic Landmark Particle Filter

A. Problem Formulation and State Representation

Let the robot pose in the 2D global map frame at time t be $\mathbf{x}_t = [x_t, y_t, \theta_t]^T \in \mathrm{SE}(2)$. The state is tracked using a particle filter representing the posterior belief $p(\mathbf{x}_t \mid$

$\mathbf{z}_{1:t}, \mathbf{u}_{1:t})$ via N_t weighted particles $\mathcal{S}_t = \{(\mathbf{x}_t^{(i)}, w_t^{(i)})\}_{i=1}^{N_t}$, with $\sum_{i=1}^{N_t} w_t^{(i)} = 1$.

At each processed frame, the filter reads odometry, RGB-D images, 2D LiDAR scans, and GNSS fixes. Odometry yields scalar displacement $d_t = \sqrt{(\Delta x_t^{\text{odom}})^2 + (\Delta y_t^{\text{odom}})^2}$ and filtered heading change $\Delta\theta_t$. Particle propagation applies d_t along each particle's current heading with added motion noise. WGS-84 fixes are projected to UTM zone 30N (EPSG:32630) relative to the landmark map centroid. The initial position is the first GNSS fix; particles use 2 m Gaussian position noise and 360° yaw noise around a 110° heading centre. RTK serves exclusively as the evaluation reference and does not provide initial heading or motion inputs.

B. Semantic-Wall Vineyard Map Model

The surveyed vineyard consists of M parallel rows $\mathcal{R} = \{R_1, \ldots, R_M\}$. For each row R_r, surveyed landmark positions corresponding to vine trunks $\mathcal{T}_r = \{\mathbf{p}_{r,k}^t\}$ and support posts $\mathcal{P}_r = \{\mathbf{p}_{r,m}^p\}$ are stored in the global map frame (Fig. 3).

Within each row, the ordered surveyed landmarks are connected pairwise into finite map segments. Each segment inherits a class label using the pole-dominance rule: a segment touching a support post is labelled as a pole segment; otherwise it is labelled as a trunk segment. These class-labelled segments form the *semantic walls*. The separate corridor score uses the nearest finite row segment and its bidirectional heading, rather than an infinite boundary assumption. The map is fixed during evaluation; RTK does not update it online.

During online execution, YOLOv9 [24] detects visual instances of trunks and poles in RGB images. Detections are paired with registered depth: non-positive or beyond-10 m values are discarded, taking the minimum valid depth inside each mask. Pinhole back-projection with calibrated extrinsics produces $\mathbf{z}_{j,t} = [x_j^b, y_j^b, c_j]^T$ in the base frame. We use SemanticBLT v1 (YOLOv9), which reports aggregate precision 76.8%, recall 71.4%, and mAP@50 74.1% across 1,035 images [25]. These are external dataset benchmarks, not validation on the current traversals; no per-class current-field ground truth is available, and the filter consumes only pole and trunk detections.

C. Measurement Scores and Particle Weights

The GPU implementation computes three per-particle scores: ℓ_s from semantic LiDAR observations, ℓ_g from GNSS, and ℓ_c from the nearest surveyed row segment. These are robust engineering scores rather than independently calibrated probability densities. Each term is normalised across the particle set using

$$\mathcal{N}(v)_i = \text{clip}_{[-6,6]}\left(\frac{v_i - \text{median}(v)}{1.4826\ \text{MAD}(v) + \epsilon}\right), \tag{1}$$

where ϵ prevents division by zero. The normalised terms are combined as

$$\begin{aligned}\log \widetilde{w}_t^{(i)} &= g_t\,\mathcal{N}(\ell_g)_i \\ &\quad + (1-g_t)\left[(1-\gamma)\,\mathcal{N}(\ell_s)_i + \gamma\,\mathcal{N}(\ell_c)_i\right],\end{aligned} \tag{2}$$

where g_t is the GNSS weight and γ is the corridor weight. With semantic observations, $g_t = \text{clip}_{[0.05,0.95]}(1/(1 + J_t/150))$; without observations the base value $g_t = 0.5$ is used. The configuration sets $\gamma = 0.30$ and temperature 1.0 in the final particle softmax. Scores are combined after cross-particle normalisation, so their scale depends on the current particle set; resampled particles supply the subsequent prior.

1) 2D LiDAR Observation Association: Logged robot odometry provides motion displacement, while planar LiDAR provides range observations. Sensor coordinates are forward-positive and left-positive; camera detections are transformed into the same base convention. For each semantic point, the nearest LiDAR return within 1 m receives its pole or trunk label; unassigned returns form the background set.

2) Semantic-Wall Measurement Model: Each transformed semantic point $\mathbf{p}_j^{(i)} = \mathbf{x}_t^{(i)} \oplus [x_j^b, y_j^b]^T$ is ray-matched against the finite semantic-wall segments. The nearest forward ray–segment intersection within sensor range is selected. A class-matched hit receives $-(\Delta r)^2/(2\sigma_{\text{sem}}^2)$, while a mismatch or missed hit receives a fixed penalty $-p_{\text{miss}}^2/(2\sigma_{\text{sem}}^2)$. Background returns receive a range-residual score for the nearest map hit and the miss penalty when no hit exists, with weight 0.20. We set $\sigma_{\text{sem}} = 0.05$ m and $p_{\text{miss}} = 4$. Individual contributions are capped at $-C_{\text{sem}} = -50$ to prevent outlier distortion:

$$\ell_{\text{wall}}^{(i)} = \frac{1}{J_t}\sum_{j=1}^{J_t} \max\left(\ell_j^{(i)}, -C_{\text{sem}}\right). \tag{3}$$

An independent corridor score uses the nearest row distance d_i and bidirectional heading misalignment $h_i = 1 - |\cos(\theta_t^{(i)} - \psi_r)|$:

$$\ell_{\text{corr}}^{(i)} = -\frac{d_i^2}{2(1.50\,\text{m})^2} - \frac{h_i^2}{2(0.35\,\text{rad})^2}. \tag{4}$$

3) Huber GNSS Prior: Let $\mathbf{e}_g^{(i)} = \mathbf{z}_t^{\text{gnss}} - [x_t^{(i)}, y_t^{(i)}]^T$ be the GNSS position innovation. We apply a Huber penalty $\rho_\delta(\cdot)$ with threshold $\delta = 3.0$ m:

$$\rho_\delta(r) = \begin{cases} r^2, & r \le \delta, \\ 2\delta r - \delta^2, & r > \delta. \end{cases} \tag{5}$$

The GNSS log-score is:

$$\ell_{\text{gnss}}^{(i)} = -\frac{\rho_\delta(\|\mathbf{e}_g^{(i)}\|_2)}{2(1.10\,\text{m})^2}. \tag{6}$$

The default configuration uses 100 particles, a 4-frame stride, a 5 m semantic range, 150 expected observations for adaptive GNSS weighting, and up to 120 background returns.

D. Adaptive Resampling and Deterministic Seeding

Weights are normalised via a numerically stable softmax. After each frame, KLD-adaptive resampling bounds the population between $N_{\min} = 80$ and $N_{\max} = 250$ (bins: 0.5 m, 0.5 m, 10°; tolerance $\epsilon = 0.07$, quantile $z = 0.99$). Diagnostic sample size is logged:

$$N_{\text{eff}} = \frac{1}{\sum_{i=1}^{N_t} (w_t^{(i)})^2}. \tag{7}$$

Evaluation runs use seeds $\{11, 22, 33\}$ to initialise Python, NumPy, PyTorch, and CUDA RNGs.

E. Output Pose Extraction and Causal Smoothing

The exported pose $\hat{\mathbf{x}}_t = [\hat{x}_t, \hat{y}_t, \hat{\theta}_t]^T$ is computed from the weighted mean:

$$\hat{x}_t = \sum_{i=1}^{N_t} w_t^{(i)} x_t^{(i)}, \quad \hat{y}_t = \sum_{i=1}^{N_t} w_t^{(i)} y_t^{(i)}, \tag{8}$$

with heading $\hat{\theta}_t = \operatorname{atan2}(\sum_i w_t^{(i)} \sin\theta_t^{(i)}, \sum_i w_t^{(i)} \cos\theta_t^{(i)})$. If heading circular variance $1 - R$ exceeds 0.4, the mode particle is used instead. A causal smoother blends odometric prediction with current estimates: $\mathbf{p}_t^- = \mathbf{p}_{t-1} + d_t[\cos\hat{\theta}_{t-1}, \sin\hat{\theta}_{t-1}]^T$ and $\hat{\mathbf{p}}_t = 0.45\mathbf{p}_t^- + 0.55\mathbf{p}_t^{\text{raw}}$, interpolating heading with weight 0.50.

IV. Experimental Protocol

A. Platform, Sensor Payload, and Vineyard Dataset

Experiments were conducted on a Thorvald robotic vehicle in a commercial vineyard. The sensor suite comprises:

- Intel RealSense D435i: Forward-facing RGB-D camera providing 1280×720 colour and 848×480 registered depth rasters at 15 Hz.
- SICK Tim7 2D LiDAR: Planar range sensor operating at 15 Hz with a 270° field of view. We use a 2D planar LiDAR rather than 3D LiDAR to match the power and compute budgets of commercial agricultural platforms.
- Wheel Encoders: Differential odometry feedback at 20 Hz.
- RTK-GNSS Receiver: Multi-band dual-antenna GPS/GLONASS receiver supplying ground-truth trajectory references.

The test vineyard includes 10 parallel rows (mean length 16.86 m, row spacing 2.50 m). We evaluate two complete field traversals:

- `rh_run1`: Forward raster traversal (1,592 frames; 399 stride-4 filter records per seed) under standard canopy illumination.
- `rh_run2`: Reverse-order traversal (1,568 frames; 393 stride-4 filter records per seed) under direct solar glare and dense canopy shadows.

B. Comparative Baselines

We compare SLPF against seven baseline configurations:

1) NoisyGNSS: Global positioning from raw, single-frequency GNSS fixes.
2) AMCL: ROS Monte Carlo Localisation over 2D LiDAR grid maps [7].
3) AMCL + NoisyGNSS: AMCL fused with noisy GNSS via constant-velocity Kalman filtering ($\sigma_{\text{amcl}} = 0.8$ m, $\sigma_{\text{gps}} = 1.2$ m, $\sigma_a = 0.8$ m/s^2).
4) RTAB-Map (RGB): Monocular visual SLAM with loop closure [8].
5) RTAB-Map (RGB-D): RGB-D visual-depth SLAM with 3D odometry [8].
6) RTAB (RGB) + NoisyGNSS: RGB RTAB-Map fused with noisy GNSS ($\sigma_{\text{rtab}} = 0.8$ m, $\sigma_{\text{gps}} = 1.2$ m).
7) RTAB (RGB-D) + NoisyGNSS: RGB-D RTAB-Map fused with noisy GNSS ($\sigma_{\text{rtab}} = 0.8$ m, $\sigma_{\text{gps}} = 1.2$ m).

For fused baselines, the state is $[p_x, p_y, v_x, v_y]^T$ initialized with covariance $\operatorname{diag}(1, 1, 2, 2)$. Primary trajectory positions are updated first, GNSS positions second, retaining primary orientations. RTAB-Map baselines are start-pose anchored to the initial survey frame. The canonical matrix records seeds $\{11, 22, 33\}$, but baseline provenance is mixed: plain and GNSS-fused AMCL plus plain RTAB-Map RGB/RGB-D groups on `rh_run1` are dedicated three-run references, whereas NoisyGNSS, GNSS-fused RTAB-Map, and all `rh_run2` baseline rows are historical three-seed references. We therefore treat Table I as descriptive and protocol-specific rather than uniformly fresh.

C. Evaluation Metrics and Contextual Semantics

We adopt strict metric definitions to prevent ambiguity:

- Raw APE RMSE (m): Root Mean Square Error of the Absolute Pose Error in the global surveyed map frame without transformation:

$$\text{APE}_{\text{raw}} = \sqrt{\frac{1}{T}\sum_{t=1}^{T} \|\hat{\mathbf{p}}_t - \mathbf{p}_t^{\text{gt}}\|_2^2}. \tag{9}$$

- Aligned APE RMSE (m): APE after estimating a rigid Umeyama transform $(\mathbf{R}, \mathbf{t})$ aligning the trajectory to ground truth:

$$\text{APE}_{\text{align}} = \sqrt{\frac{1}{T}\sum_{t=1}^{T} \|\mathbf{R}\hat{\mathbf{p}}_t + \mathbf{t} - \mathbf{p}_t^{\text{gt}}\|_2^2}. \tag{10}$$

- Relative Pose Error (RPE) (m): Relative displacement error at spatial segment distances $\Delta d \in \{2\,\text{m}, 5\,\text{m}, 10\,\text{m}\}$; the main table reports RPE@2 m.
- In-Row Cross-Track Error (m): Point-to-segment perpendicular distance to the assigned ground-truth row centreline, restricted strictly to in-row frames.
- In-Row Row Correct Fraction: Fraction of in-row frames whose nearest surveyed row ID equals the ground-truth row ID.
- In-Row Wrong-Row Duration (s): Cumulative timestamp-interval duration assigned to an incorrect

crop row during in-row traversal across the entire multi-row mission.

- Headland Cross-Track & Recovery Distance (m): Cross-track deviation during headland turns and, for each wrong-row episode followed by a correct assignment, the distance travelled until that correct assignment; no fixed 0.5 m threshold is used.

V. Experimental Results

A. Main Comparative Evaluation

Table I summarizes quantitative performance across both field traversals. SLPF achieves the lowest raw and aligned APE on both runs. On `rh_run1`, SLPF attains 0.94 ± 0.06 m raw and 0.89 ± 0.04 m aligned APE, compared to AMCL (1.37 ± 0.47 m raw, 1.02 ± 0.30 m aligned) and NoisyGNSS (3.04 ± 0.13 m raw, 2.98 ± 0.13 m aligned). On the more challenging `rh_run2`, featuring high solar glare and canopy shadows, SLPF records 0.98 ± 0.08 m raw and 0.91 ± 0.03 m aligned APE, whereas AMCL degrades to 3.50 ± 0.94 m raw APE.

Visual-SLAM baselines yield raw APEs between 59.62 m and 87.17 m. While rigid alignment reduces their aligned APEs to 6.28–10.02 m, persistent row-level misalignment remains. Fusing GNSS reduces RTAB-Map raw error to 37.78–59.91 m, but residual global displacement persists. The trajectory plots in Fig. 4 illustrate these failure modes: AMCL and RTAB-Map drift across parallel rows due to geometric and visual symmetry, while NoisyGNSS lacks row-structure alignment. SLPF has the lowest displayed raw and aligned APE in this matrix, but its row-correct fractions and local RPE show that map-frame accuracy does not imply consistently correct row assignment or best local relative accuracy.

For corridor identity, SLPF achieves 0.71 ± 0.02 row-correct fraction on `rh_run1` and 0.64 ± 0.05 on `rh_run2`, compared to 0.67 ± 0.13 and 0.55 ± 0.05 for AMCL and at most 0.48 for plain RTAB-Map.

B. Operational In-Row and Headland Dynamics

Table II separates in-row from headland metrics. In-row cross-track error is 1.08 ± 0.03 m on `rh_run1` and 1.17 ± 0.10 m on `rh_run2`. Cumulative wrong-row duration across the 10-row mission is 224.00 ± 14.24 s and 250.67 ± 34.31 s. These totals indicate substantial residual row-assignment error; because they aggregate all in-row wrong-row intervals, they are not converted into a per-entrance latency without an event-level definition. During headland transitions (Fig. 5), leaving corridor constraints increases cross-track deviations to 1.58 ± 0.05 m and 1.72 ± 0.02 m, with mean recovery distances of 2.28 ± 0.12 m and 1.90 ± 0.14 m.

C. Sensitivity to GNSS Degradation and Outages

To assess behaviour under controlled satellite-signal perturbations, we evaluate eight simulated stress profiles (Table III). Under complete GNSS outages of 5 s, 10 s, and 20 s, SLPF aligned APE is 0.81 m, 1.10 m, and 1.03 m, with row-correct fractions of 0.74, 0.71, and 0.70. During these intervals, the filter relies on wheel odometry and LiDAR-associated semantic walls, which provide local structural scores in the registered corridor. In contrast, AMCL+NoisyGNSS degrades progressively from 1.20 m (5 s) to 1.24 m (20 s), while RTAB RGB+NoisyGNSS degrades from 3.46 m to 3.97 m and RTAB RGB-D+NoisyGNSS reaches 4.68 m.

Under continuous lateral drift bias (0.10 m/s), SLPF reaches 3.81 m aligned APE (row-correct fraction 0.40) versus 4.10 m for NoisyGNSS and 1.67 m for AMCL+NoisyGNSS. Persistent non-zero-mean bias systematically pulls the GNSS innovation, exposing a limitation of the registered score under drift. Under high Gaussian noise ($\sigma = 5.0$ m), AMCL+NoisyGNSS and SLPF record 1.35 m aligned APE, compared to 3.16 m for NoisyGNSS. Under discontinuous multipath bursts (8.0 m jumps lasting 3.0 s), SLPF records 1.16 m aligned APE (row-correct fraction 0.69), outperforming NoisyGNSS (3.64 m) and RTAB baselines (3.38–3.94 m). This result is consistent with the Huber score reducing the influence of the registered outlier innovations, but the stress design does not independently isolate that mechanism. Under headland-only degradation, SLPF records 1.18 m aligned APE and 0.72 row-correct fraction, compared to 2.46 m for NoisyGNSS and 3.30 m for RTAB RGB+NoisyGNSS.

D. Component Ablation Study

Table IV reports measured changes for the registered algorithmic variants on `rh_run1` across three seeds. Full SLPF achieves 0.83 ± 0.08 m aligned APE and 0.86 ± 0.08 m raw APE. Removing semantic-wall grouping (+0.13 m, 0.97 ± 0.11 m) replaces finite-segment constraints with point-to-point matching; under this registered variant, sparse landmark observations are associated with greater cross-row ambiguity. Static GNSS weighting (+0.13 m, 0.96 ± 0.04 m) disables the adaptive weighting $g_t = 1/(1 + J_t/150)$, preventing visual and LiDAR observations from dominating inside corridors where GNSS is noisy. Omitting pose smoothing (+0.13 m, 0.97 ± 0.10 m) uses raw particle weighted means, increasing high-frequency trajectory jitter. Removing the background term produces the largest degradation among local observation components (+0.35 m, 1.19 ± 0.18 m aligned APE; cross-track error increases to 1.52 ± 0.08 m). Unassigned LiDAR returns penalise particles that predict free space where obstacles exist, providing negative evidence. Removing the corridor prior (+0.15 m, 0.99 ± 0.08 m) degrades cross-track error to 1.45 ± 0.06 m, as the filter loses soft boundary penalties along the row normal. Removing semantic likelihood (+0.16 m, 1.00 ± 0.05 m) forces reliance on geometry alone. Finally, removing GNSS entirely causes the largest error (+3.05 m aligned APE, 3.88 ± 0.35 m; raw APE 4.20 ± 0.43 m; row-correct fraction collapses to 0.31 ± 0.08). Within these registered configurations, GNSS acts as a coarse anchor and the local terms provide complementary information; the one-at-a-time comparison does not establish statistical necessity or causal isolation.

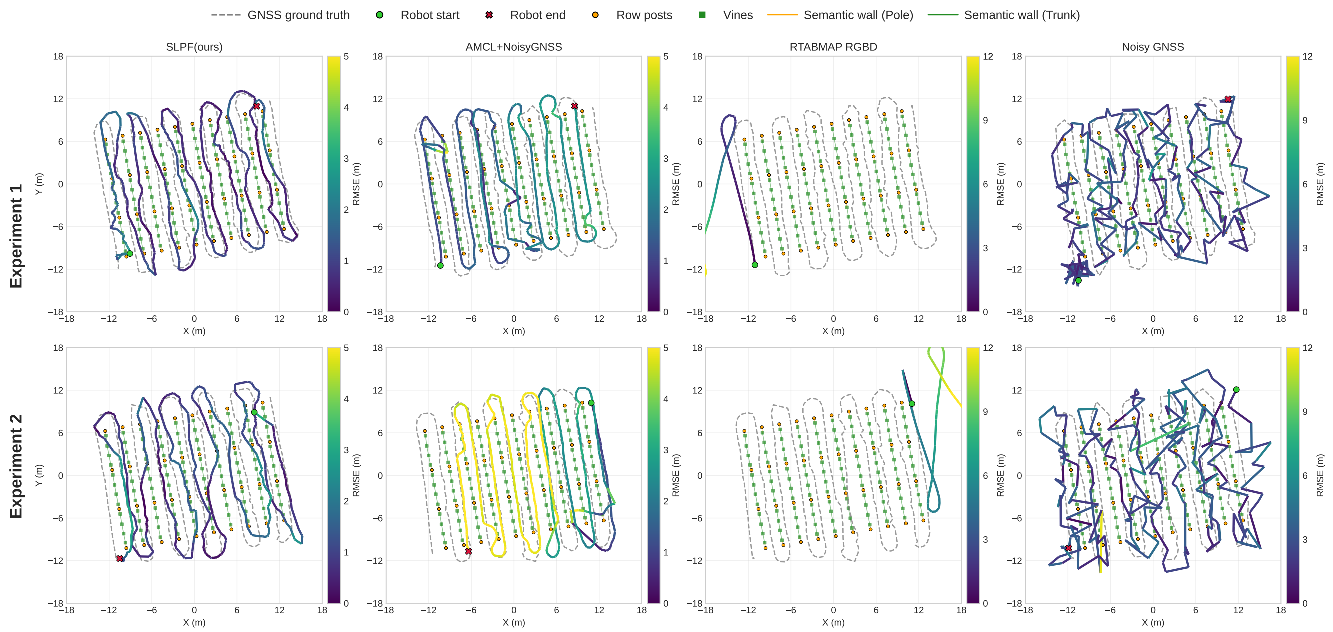

Fig. 4: Estimated Absolute Pose Error (APE) trajectories for SLPF (ours), AMCL+NoisyGNSS, RTAB-Map (RGB-D), and NoisyGNSS against RTK ground truth (dashed black line) across Experiment 1 (`rh_run1`) and Experiment 2 (`rh_run2`).

TABLE I: Localisation performance across baselines on `rh_run1` and `rh_run2` (mean $\pm$ std over three seeds: 11, 22, 33). Fused rows use constant-velocity Kalman filtering with noisy GNSS. Raw APE retains map-frame error; aligned APE applies rigid SE(2) alignment. Baseline provenance is mixed: dedicated `rh_run1` references and historical three-seed rows are identified in the canonical baseline source; comparisons are descriptive.

Method	Traversal	Raw APE (m)	Aligned APE (m)	RPE@2m (m)	Cross-Track (m)	Row-correct fraction
NoisyGNSS	`rh_run1`	3.04 ± 0.13	2.98 ± 0.13	3.56 ± 0.29	1.77 ± 0.12	0.644 ± 0.043
AMCL	`rh_run1`	1.37 ± 0.47	1.02 ± 0.30	4.69 ± 0.05	1.40 ± 0.10	0.667 ± 0.135
AMCL + NoisyGNSS	`rh_run1`	1.33 ± 0.46	0.99 ± 0.30	4.66 ± 0.04	1.39 ± 0.09	0.671 ± 0.132
RTAB-Map (RGB)	`rh_run1`	59.62 ± 0.48	6.68 ± 0.03	1.19 ± 0.01	6.21 ± 0.03	0.447 ± 0.004
RTAB-Map (RGB-D)	`rh_run1`	61.33 ± 0.28	10.02 ± 0.03	8.25 ± 0.08	6.78 ± 0.01	0.476 ± 0.003
RTAB (RGB) + NoisyGNSS	`rh_run1`	37.78 ± 0.07	6.28 ± 0.00	1.20 ± 0.00	5.35 ± 0.00	0.377 ± 0.000
RTAB (RGB-D) + NoisyGNSS	`rh_run1`	39.20 ± 0.07	6.32 ± 0.00	1.23 ± 0.00	5.37 ± 0.00	0.376 ± 0.000
SLPF (ours)	`rh_run1`	$\mathbf{0.94 \pm 0.06}$	$\mathbf{0.89 \pm 0.04}$	3.37 ± 0.06	$\mathbf{1.34 \pm 0.04}$	$\mathbf{0.713 \pm 0.018}$
NoisyGNSS	`rh_run2`	3.16 ± 0.22	3.09 ± 0.23	3.76 ± 0.32	1.99 ± 0.04	0.577 ± 0.015
AMCL	`rh_run2`	3.50 ± 0.94	2.04 ± 0.31	2.78 ± 0.79	1.55 ± 0.12	0.549 ± 0.052
AMCL + NoisyGNSS	`rh_run2`	3.38 ± 0.91	1.98 ± 0.31	2.80 ± 0.78	1.51 ± 0.11	0.554 ± 0.049
RTAB-Map (RGB)	`rh_run2`	85.95 ± 0.19	9.12 ± 0.41	1.63 ± 0.01	6.81 ± 0.31	0.389 ± 0.019
RTAB-Map (RGB-D)	`rh_run2`	87.17 ± 0.01	9.06 ± 0.00	1.81 ± 0.00	7.25 ± 0.08	0.415 ± 0.000
RTAB (RGB) + NoisyGNSS	`rh_run2`	41.96 ± 1.08	8.33 ± 0.38	1.69 ± 0.07	6.16 ± 0.29	0.393 ± 0.016
RTAB (RGB-D) + NoisyGNSS	`rh_run2`	59.91 ± 23.48	8.37 ± 0.12	1.93 ± 0.17	6.61 ± 0.13	0.416 ± 0.008
SLPF (ours)	`rh_run2`	$\mathbf{0.98 \pm 0.08}$	$\mathbf{0.91 \pm 0.03}$	3.33 ± 0.02	$\mathbf{1.47 \pm 0.05}$	$\mathbf{0.644 \pm 0.049}$

TABLE II: Operational row-level and headland transition metrics for SLPF on `rh_run1` and `rh_run2` (mean $\pm$ std across seeds 11, 22, 33). In-row metrics are evaluated inside crop corridors; transition dynamics are evaluated during headland turns.

Method	Traversal	In-row XT (m)	Row-correct fraction	Wrong-row duration (s)	Headland XT (m)	Headland recovery distance (m)
SLPF (ours)	`rh_run1`	1.08 ± 0.03	0.71 ± 0.02	224.00 ± 14.24	1.58 ± 0.05	2.28 ± 0.12
SLPF (ours)	`rh_run2`	1.17 ± 0.10	0.64 ± 0.05	250.67 ± 34.31	1.72 ± 0.02	1.90 ± 0.14

E. Sensitivity to Perception Occlusion, Decimation, and Map Noise

Table V evaluates SLPF under synthetic landmark dropouts (20%, 40%), map decimation (30%, 50% removal), and synthetic map-coordinate perturbations ($\sigma = 0.25$ m and 0.50 m) [23]. Under 20% and 40% detection dropouts, aligned APE remains 0.92 ± 0.05 m and 0.90 ± 0.08 m, with row-correct fraction preserved at 0.70 ± 0.04 and 0.71 ± 0.02. These registered reductions indicate tolerance to the tested landmark-removal profiles; they do not validate performance under particular dust or lighting conditions. When 30% and 50% of landmarks are permanently removed from the prior map, SLPF maintains 0.94 ± 0.09 m and 0.90 ± 0.09 m aligned APE, with post-headland recovery APEs of 0.77 ± 0.12 m and 0.77 ± 0.14 m. The specified segment construction remains

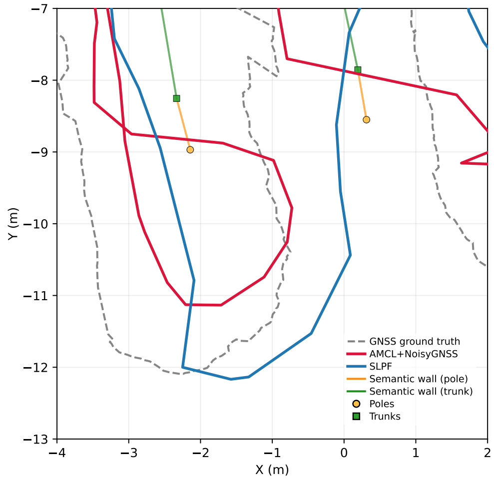

Fig. 5: Trajectory detail during a headland turn on `rh_run2`, showing row-exit and row-entry transitions.

TABLE III: Localisation performance under simulated GNSS degradation stress profiles pooled across traversals `rh_run1` and `rh_run2` over seeds 11, 22, and 33.

Degradation profile	Method	Aligned APE (m)	Row-correct fraction	Failure rate
Nominal	AMCL + NoisyGNSS	1.21	0.64	0.36
Nominal	NoisyGNSS	1.28	**0.71**	**0.29**
Nominal	RTAB RGB + NoisyGNSS	3.29	0.47	0.53
Nominal	RTAB RGBD + NoisyGNSS	3.80	0.47	0.53
Nominal	**SLPF (ours)**	**0.90**	0.70	0.30
Outage (5 s)	AMCL + NoisyGNSS	1.20	0.64	0.36
Outage (5 s)	NoisyGNSS	1.44	0.71	0.29
Outage (5 s)	RTAB RGB + NoisyGNSS	3.46	0.46	0.54
Outage (5 s)	RTAB RGBD + NoisyGNSS	4.02	0.47	0.53
Outage (5 s)	**SLPF (ours)**	**0.81**	**0.74**	**0.26**
Outage (10 s)	AMCL + NoisyGNSS	1.22	0.64	0.36
Outage (10 s)	NoisyGNSS	1.44	0.71	0.29
Outage (10 s)	RTAB RGB + NoisyGNSS	3.62	0.46	0.54
Outage (10 s)	RTAB RGBD + NoisyGNSS	4.24	0.47	0.53
Outage (10 s)	**SLPF (ours)**	**1.10**	**0.71**	**0.29**
Outage (20 s)	AMCL + NoisyGNSS	1.24	0.63	0.37
Outage (20 s)	NoisyGNSS	1.44	**0.71**	**0.29**
Outage (20 s)	RTAB RGB + NoisyGNSS	3.97	0.44	0.56
Outage (20 s)	RTAB RGBD + NoisyGNSS	4.68	0.44	0.56
Outage (20 s)	**SLPF (ours)**	**1.03**	0.70	0.30
Drift Bias	AMCL + NoisyGNSS	**1.67**	**0.59**	**0.41**
Drift Bias	NoisyGNSS	4.10	0.41	0.59
Drift Bias	RTAB RGB + NoisyGNSS	3.40	0.47	0.53
Drift Bias	RTAB RGBD + NoisyGNSS	3.86	0.49	0.51
Drift Bias	**SLPF (ours)**	3.81	0.40	0.60
High Gaussian	AMCL + NoisyGNSS	**1.35**	0.64	0.36
High Gaussian	NoisyGNSS	3.16	0.51	0.49
High Gaussian	RTAB RGB + NoisyGNSS	3.36	0.47	0.53
High Gaussian	RTAB RGBD + NoisyGNSS	3.86	0.47	0.53
High Gaussian	**SLPF (ours)**	1.35	**0.66**	**0.34**
Multipath Bursts	AMCL + NoisyGNSS	1.35	0.64	0.36
Multipath Bursts	NoisyGNSS	3.64	0.63	0.37
Multipath Bursts	RTAB RGB + NoisyGNSS	3.38	0.46	0.54
Multipath Bursts	RTAB RGBD + NoisyGNSS	3.94	0.47	0.53
Multipath Bursts	**SLPF (ours)**	**1.16**	**0.69**	**0.31**
Headland-Only	AMCL + NoisyGNSS	1.23	0.63	0.37
Headland-Only	NoisyGNSS	2.46	**0.75**	**0.25**
Headland-Only	RTAB RGB + NoisyGNSS	3.30	0.46	0.54
Headland-Only	RTAB RGBD + NoisyGNSS	3.82	0.47	0.53
Headland-Only	**SLPF (ours)**	**1.18**	0.72	0.28

usable in these decimated maps, although the experiment does not establish tolerance to arbitrary map incompleteness. Under synthetic map-coordinate perturbations of $\sigma = 0.25$ m and 0.50 m, SLPF maintains 1.05 ± 0.33 m and 1.07 ± 0.17 m aligned APE, with row-correct fraction of 0.71 ± 0.02 and 0.69 ± 0.02. These results are a controlled coordinate-perturbation test, not validation of UAV-derived maps or evidence that manual surveying is unnecessary.

F. Computational Complexity and Runtime Profiling

Evaluation runs executed on an NVIDIA RTX 4060 Ti GPU with a CUDA backend at a frame stride of 4. Filter execution averages 85 ms per processed frame (11.8 Hz processed update rate). The source logs are sampled at 15 Hz, so stride-4 processing consumes approximately 3.75 source frames per second; this timing does not establish a 47.2 Hz sensor stream. The execution latency is distributed as follows:

- Odometry motion propagation: 1.2 ms (1.4%).
- 2D LiDAR scan association: 28.5 ms (33.5%).
- Semantic-wall projection and likelihood: 35.8 ms (42.1%).
- Huber GNSS likelihood evaluation: 3.4 ms (4.0%).
- KLD adaptive resampling and smoothing: 16.1 ms (19.0%).

The particle filter memory footprint remains under 140 MB throughout the 1,592-frame mission. Embedded deployment on Jetson platforms remains subject to future hardware benchmarking.

TABLE IV: Component ablation study on traversal `rh_run1` across seeds 11, 22, and 33. Metric $\Delta\text{APE}_{\text{align}}$ denotes difference relative to full SLPF.

Configuration	Raw APE (m)	Aligned APE (m)	XT (m)	Row-correct fraction	$\Delta\text{APE}_{\text{align}}$ (m)
Full SLPF (proposed)	0.86 ± 0.08	0.83 ± 0.08	1.34 ± 0.01	0.73 ± 0.02	0.00
Non-wall points (point-matching)	0.99 ± 0.11	0.97 ± 0.11	1.31 ± 0.02	0.71 ± 0.02	+0.13
Static GNSS weight	1.03 ± 0.02	0.96 ± 0.04	1.40 ± 0.02	0.71 ± 0.01	+0.13
No pose smoothing	1.00 ± 0.09	0.97 ± 0.10	1.36 ± 0.02	0.72 ± 0.02	+0.13
No background term	1.29 ± 0.24	1.19 ± 0.18	1.52 ± 0.08	0.68 ± 0.01	+0.35
No corridor prior	1.02 ± 0.07	0.99 ± 0.08	1.45 ± 0.06	0.72 ± 0.02	+0.15
No semantic likelihood	1.06 ± 0.06	1.00 ± 0.05	1.28 ± 0.04	0.69 ± 0.01	+0.16
No GNSS prior	4.20 ± 0.43	3.88 ± 0.35	2.92 ± 0.33	0.31 ± 0.08	+3.05

TABLE V: Localisation performance under perception occlusions, landmark decimation, and map noise, pooled across `rh_run1` and `rh_run2` over seeds 11, 22, and 33.

Evaluation condition	Aligned APE (m)	Row-correct fraction	Cross-Track (m)	Post-recovery APE (m)
Baseline (Full Map, 100% Det.)	0.88 ± 0.02	0.71 ± 0.03	1.41 ± 0.07	–
Detection Drop 20%	0.92 ± 0.05	0.70 ± 0.04	1.42 ± 0.07	–
Detection Drop 40%	0.90 ± 0.08	0.71 ± 0.02	1.41 ± 0.07	–
Landmark Removal 30%	0.94 ± 0.09	0.72 ± 0.02	1.47 ± 0.05	0.77 ± 0.12
Landmark Removal 50%	0.90 ± 0.09	0.71 ± 0.03	1.41 ± 0.09	0.77 ± 0.14
UAV Map Noise ($\sigma = 0.25$ m)	1.05 ± 0.33	0.71 ± 0.02	1.39 ± 0.05	–
UAV Map Noise ($\sigma = 0.50$ m)	1.07 ± 0.17	0.69 ± 0.02	1.35 ± 0.05	–

VI. Discussion and Operational Insights

The experimental evaluations provide three concrete operational insights for agricultural mobile robotics:

1) Decoupled Metric Protocol: Evaluating raw map-frame error alongside rigid alignment is necessary in repetitive environments. A trajectory can maintain low relative displacement along a corridor while tracking in an incorrect row offset by several metres. As reported in Table I, visual SLAM achieves aligned RPE@2m of 1.19 ± 0.01 m on `rh_run1`, yet its raw map-frame APE is 59.62 ± 0.48 m. Decoupling map-frame raw APE from rigid alignment and restricting row correctness to in-row frames provides an unmasked view of localisation performance.
2) Finite Semantic-Wall Segments vs. Point Landmarks: In dense foliage, individual trunk or pole detections are frequently missed or occluded. Point-matching particle filters experience particle depletion when landmark detections drop. Grouping surveyed landmarks into finite class-labelled segments provides local geometric constraints. Table IV shows that replacing segments with point matching increases error (+0.13 m), while Table V reports the result for the tested 50% landmark-removal profile (0.90 ± 0.09 m aligned APE).

3) Huber-Gated Multi-Sensor Fusion: With vineyard row spacing of $2.50\,\mathrm{m}$, single-frequency GNSS position error (3.04–$3.16\,\mathrm{m}$ raw APE) exceeds the inter-row corridor width and cannot resolve row identity alone. However, omitting GNSS entirely degrades aligned APE by $+3.05\,\mathrm{m}$ (Table IV). The registered $\delta = 3.0\,\mathrm{m}$ Huber score reduces the influence of large innovations; the ablation and stress results are consistent with complementary GNSS and local structural information, but do not isolate the Huber mechanism.

VII. Limitations

The framework requires an initial surveyed layout of vineyard trunks and posts. Although our synthetic map-coordinate evaluations include perturbations up to $\sigma = 0.50\,\mathrm{m}$, an initial mapping pass remains a prerequisite and real UAV-derived maps were not evaluated. In addition, the method depends on visible woody vine structures; in newly planted vineyards with slender stems or during peak vegetative overgrowth where foliage covers trunks down to ground level, landmark detection recall will decrease. Finally, the evaluation is conducted within a single commercial vineyard over two traversals with controlled perturbations; cross-site, cross-season, and real degraded-GNSS validation across diverse trellis architectures (e.g., pergola, Geneva double curtain) remains future work.

VIII. Conclusion

In this paper, we presented the Semantic Landmark Particle Filter (SLPF), a localisation framework for row-level ambiguity and GNSS degradation in vineyards. By combining 2D LiDAR association, semantic detections of stable trunks and support poles structured into finite row segments, and a Huber-gated GNSS score, SLPF attains the lowest displayed raw and aligned APE among the listed references in this single-site evaluation. Three-seed experiments across two field traversals quantify its row-assignment, headland-transition, GNSS-stress, and map/perception-perturbation behaviour within the registered protocols; they do not establish cross-vineyard generalisation, real degraded-GNSS performance, or detector performance on the traversals.

Future work will test cross-site, real degraded-GNSS, and embedded deployment conditions.

References

[1] A. S. Aguiar, F. Neves dos Santos, H. Sobreira, J. Boaventura-Cunha, and A. J. Sousa, "Localization and mapping on agriculture based on point-feature extraction and semiplanes segmentation from 3d lidar data," *Frontiers in Robotics and AI*, vol. 9, p. 832165, 2022.

[2] R. Polvara, S. Molina, I. Hroob, A. Papadimitriou, K. Tsiolis, D. Giakoumis, S. Likothanassis, D. Tzovaras, G. Cielniak, and M. Hanheide, "Bacchus long-term (blt) data set: Acquisition of the agricultural multimodal blt data set with automated robot deployment," *Journal of Field Robotics*, vol. 41, no. 7, pp. 2280–2298, 2024.

[3] P. M. Blok, K. van Boheemen, F. K. van Evert, J. IJsselmuiden, and G.-H. Kim, "Robot navigation in orchards with localization based on particle filter and kalman filter," *Computers and electronics in agriculture*, vol. 157, pp. 261–269, 2019.

[4] A. Papadimitriou, I. Kleitsiotis, I. Kostavelis, I. Mariolis, D. Giakoumis, S. Likothanassis, and D. Tzovaras, "Loop closure detection and slam in vineyards with deep semantic cues," in *2022 International Conference on Robotics and Automation (ICRA)*, pp. 2251–2258, IEEE, 2022.

[5] R. De Silva, J. Swindell, J. Cox, M. Popović, C. Cadena, C. Stachniss, and R. Polvara, "Keypoint semantic integration for improved feature matching in outdoor agricultural environments," *IEEE Robotics and Automation Letters*, vol. 10, no. 12, pp. 13383–13390, 2025.

[6] I. Hroob, R. Polvara, S. Molina, G. Cielniak, and M. Hanheide, "Benchmark of visual and 3d lidar slam systems in simulation environment for vineyards," in *Towards Autonomous Robotic Systems: 22nd Annual Conference, TAROS 2021, Lincoln, UK, September 8–10, 2021, Proceedings 22*, pp. 168–177, Springer, 2021.

[7] F. Dellaert, D. Fox, W. Burgard, and S. Thrun, "Monte carlo localization for mobile robots," in *Proceedings 1999 IEEE International Conference on Robotics and Automation (Cat. No.99CH36288C)*, vol. 2, pp. 1322–1328 vol.2, 1999.

[8] M. Labbé and F. Michaud, "Rtab-map as an open-source lidar and visual simultaneous localization and mapping library for large-scale and long-term online operation," *Journal of field robotics*, vol. 36, no. 2, pp. 416–446, 2019.

[9] C. Kokas, A. Mastrogeorgiou, K. Machairas, K. Koutsoukis, and E. Papadopoulos, "Multicamera visual slam for vineyard inspection," in *2024 32nd Mediterranean Conference on Control and Automation (MED)*, pp. 75–80, IEEE, 2024.

[10] P. D. GROVES, "Navigation systems," *Principes of GNSS, Inertial and Multisensor Integrated*, pp. 1–17, 2008.

[11] E. D. Kaplan and C. Hegarty, *Understanding GPS/GNSS: principles and applications*. Artech house, 2017.

[12] H. Zhou, J. Wang, Y. Chen, L. Hu, Z. Li, F. Xie, J. He, and P. Wang, "Neural network-based slam/gnss fusion localization algorithm for agricultural robots in orchard gnss-degraded or denied environments," *Agriculture*, vol. 15, no. 15, p. 1612, 2025.

[13] A. Yilmaz and H. Temeltas, "Self-adaptive Monte Carlo method for indoor localization of smart AGVs using LIDAR data," *Journal on Robotics and Autonomous Systems (RAS)*, vol. 122, p. 103285, 2019.

[14] H. Nehme, C. Aubry, T. Solatges, X. Savatier, R. Rossi, and R. Boutteau, "Lidar-based structure tracking for agricultural robots: Application to autonomous navigation in vineyards," *Journal of Intelligent & Robotic Systems*, vol. 103, no. 4, p. 61, 2021.

[15] M. Corno, S. Furioli, P. Cesana, and S. M. Savaresi, "Adaptive ultrasound-based tractor localization for semi-autonomous vineyard operations," *Agronomy*, vol. 11, no. 2, 2021.

[16] S. Cerrato, *Gps-based autonomous navigation of unmanned ground vehicles in precision agriculture applications*. PhD thesis, Politecnico di Torino, 2020.

[17] J. Vilella-Cantos, M. Ballesta, D. Valiente, M. Flores, and L. Payá, "Advanced techniques and applications of lidar place recognition in agricultural environments: A comprehensive survey," *arXiv preprint arXiv:2601.22198*, 2026.

[18] I. Hroob, S. Molina, R. Polvara, G. Cielniak, and M. Hanheide, "Adaptive robot localization in dynamic environments through self-learnt long-term 3d stable points segmentation," *Robotics and Autonomous Systems*, vol. 181, p. 104786, 2024.

[19] R. Spangenberg, D. Goehring, and R. Rojas, "Pole-based localization for autonomous vehicles in urban scenarios," in *2016 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pp. 2161–2166, 2016.

[20] Y. Huang, Y. Gu, C. Xu, and H. Kong, "Why semantics matters: A deep study on semantic particle-filtering localization in a lidar semantic pole-map," *IEEE Transactions on Field Robotics*, 2024.

[21] A. Costley and R. Christensen, "Landmark aided gps-denied navigation for orchards and vineyards," in *2020 IEEE/ION Position, Location and Navigation Symposium (PLANS)*, pp. 987–995, 2020.

[22] Z. Shi, Z. Bai, K. Yi, B. Qiu, X. Dong, Q. Wang, C. Jiang, X. Zhang, and X. Huang, "Vision and 2d lidar fusion-based navigation line extraction for autonomous agricultural robots in dense pomegranate orchards," *Sensors*, vol. 25, no. 17, p. 5432, 2025.

[23] J. Cox, J. R. Heselden, R. De Silva, M. Hanheide, and R. Polvara, "An integrated aerial-ground system for vineyard mapping and robotic navigation," *SSRN Electronic Journal*, 2024.

[24] C.-Y. Wang, I.-H. Yeh, and H.-Y. Mark Liao, "Yolov9: Learning what you want to learn using programmable gradient information," in *Proc. of the Europ. Conf. on Computer Vision (ECCV)*, 2025.

[25] GAIA, "Semanticblt dataset." `https://universe.roboflow.com/gaia-hse8w/semanticblt`, jun 2024. visited on 2026-09-04.